

Lorentz equation

- Forces felt on a charged particle (proton in this case) moving in a uniform magnetic field
- mag field pointing up in z direction

newtons 2nd for charged particle

$$\vec{F} = m\vec{a} = q(\vec{v} \times \vec{B})$$

Lorentz force: $q(\vec{v} \times \vec{B})$

charge
(proton = 1.6×10^{-19} coulombs)

velo
vector
of particle

mag field
vector

$$F = q(\vec{v} \times \vec{B})$$

$$\vec{v} \times \vec{B} = \begin{pmatrix} v_y B_z - v_z B_y \\ v_z B_x - v_x B_z \\ v_x B_y - v_y B_x \end{pmatrix}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{q}{m}(\vec{v} \times \vec{B})$$

$$\frac{q}{m} \text{ (charge to mass ratio)} = 9.58 \times 10^{-8} \text{ C/Kg (proton)}$$

$$a_x = \frac{dv_x}{dt} \quad a_y = \frac{dv_y}{dt} \quad a_z = \frac{dv_z}{dt}$$

$$\vec{B} = (0, 0, B_z)$$

$$\vec{v} \times \vec{B} = \begin{pmatrix} v_y B_z - v_z(0) \\ v_z(0) - v_x B_z \\ v_x(0) - v_y(0) \end{pmatrix} = \begin{pmatrix} v_y B_z \\ -v_x B_z \\ 0 \end{pmatrix}$$

$$\frac{dv_x}{dt} = \frac{q}{m} v_y B_z$$

$$\frac{dv_y}{dt} = -\frac{q}{m} v_x B_z$$

$$\frac{dv_z}{dt} = 0$$

cyclotron frequency / gyrofrequency

$$\Omega = \frac{qB_z}{m} \quad (\text{rad/s})$$

$$\text{so, } \frac{dv_x}{dt} = \Omega v_y$$

$$\frac{dv_y}{dt} = -\Omega v_x$$

$$\frac{dv_z}{dt} = 0$$

diff again

$$\frac{d^2 v_x}{dt^2} = \Omega \frac{dv_y}{dt}$$

$$\frac{d^2 v_x}{dt^2} = \Omega (-\Omega v_x) = -\Omega^2 v_x$$

$$v_x(t) = A \cos(\Omega t) + B \sin(\Omega t)$$

$$v_0(t=0) = (v_x, 0, 0)$$

$$\text{@ } t=0 \quad v_x(0) = A \cos(\Omega \cdot 0) + B \sin(\Omega \cdot 0)$$

$$v_x = v_0$$

$$v_0 = A(1) + B(0)$$

$$A = v_0$$

$$\frac{dv_x}{dt} = \Omega v_y \quad \Big|_{t=0} \quad \Omega v_y(0) = 0, \quad \frac{dv_x}{dt} = 0$$

$$\frac{dv_x}{dt} = -A\Omega \sin(\Omega t) + B\Omega \cos(\Omega t)$$

$$\begin{aligned} @ t=0 \quad & -A\Omega \sin(0) + B\Omega \cos(0) \\ & = 0 + B\Omega = 0 \end{aligned}$$

So B must $= 0$

$$v_x(t) = A \cos(\Omega t) + B \sin(\Omega t)$$

$$v_x(t) = v_0 \cos(\Omega t)$$

remember: $\frac{dv_x}{dt} = \Omega v_y$

$$-v_0 \sin(\Omega t) = \Omega v_y$$

$$v_y(t) = -v_0 \sin(\Omega t)$$

$$v_z(t) = 0 \quad (\text{never changes})$$

position:

$$x(t) = \int v_0 \cos(\Omega t) dt = \frac{v_0}{\Omega} \sin(\Omega t) + x_0$$

$$y(t) = \int -v_0 \sin(\Omega t) dt = \frac{v_0}{\Omega} \cos(\Omega t) + y_0$$

$$z(t) = \int 0 dt = \text{constant}$$

motion

$\sin$ & $\cos$ same amp, parametric eq of a circle

$$x^2 + y^2 = \left(\frac{v_0}{\Omega}\right)^2$$

radius of said circle

$$r_g = \frac{v_0}{\Omega} = \frac{mv_0}{qB_x}$$

↳ gyro radius

- heavier particle = bigger circle
- faster " "
- Stronger mag field = smaller circle

Conclusion

- proton in uniform mag field traces perfect circle in plane $\perp$ to B
- rad of that circle: $r = \frac{mv_0}{qB_z}$
- Motion parallel to B unaffected